

Alejandro Enriquez

Embedded Software Engineer

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Summary

Embedded Software Engineer with 4 years developing firmware on CAN-based multi-ECU architectures across refrigeration control and off-highway vehicle systems. Proven ability to own problems end-to-end, from design through validation and manufacturing support. Delivered a cross-generation runtime software health monitoring system, reduced build times by ~65%, and diagnosed and unblocked stalled production lines. Experienced in RTOS environments (embOS, MQX), V-model, HIL testing, and cross-site collaboration.

Key Achievements

- 65% faster builds (20+ min to 7 min)
- 2 production-critical products diagnosed and unblocked
- Runtime software health monitor (STW) deployed across 2 hardware generations: 16 tasks monitored with CAN fault diagnostics

Professional Experience

Firmware & Embedded Software Engineer — *Carrier Transicold | Monterrey Technology Hub* | May 2025 – Present

- Established the Monterrey site's embedded software capability from zero as the first Transicold firmware engineer: built testbenches, tooling, and documentation infrastructure independently.
- Developed and validated firmware following a V-model process on a multi-ECU refrigeration control platform, covering ARM Cortex-M4 (embOS) and ARM Cortex-M7 (MQX) targets.
- Designed and implemented a runtime software health monitoring system (STW) for two hardware generations (ARM Cortex-M4/embOS, ARM Cortex-M7/MQX), iterating from a timestamp-based design to a counter-based approach after identifying performance issues and false alarms caused by critical sections. Features: 16-task monitoring, auto-registering task API, per-task counters, fault injection testing API, and CAN-bus diagnostic messages with task ID, ASCII identifier, and missed-count reporting.
- Traveled to Syracuse, NY to train on testbench setup and perform initial STW hardware validation, then equipped the Monterrey site with CAN sniffers, J-Link debuggers, and ECU boards to enable on-site testing capability.

- Built the site's first physical testbench from scratch, validating 5+ ECUs across 2 product generations.
- Diagnosed and unblocked two refrigeration products stalled for months in PPAP: traced one failure to a missing CAN activation sequence in manufacturing test scripts; the other to incorrect firmware being flashed due to board version confusion and an outdated Windchill PLM configuration.
- Reduced build time by ~65% (20+ min to 7 min) by developing a Python automation script replacing a multi-IDE, multi-step manual compilation process. In active use by the team; release of v1.1 in progress.
- Performed debugging and validation using J-Link, Kvaser, and Peak, including CAN/J1939 bus analysis, real-time diagnostics, and issue isolation across multi-ECU systems.

Embedded Software Engineer — *Terex Corporation* | *Monterrey, Nuevo León* | *Mar 2023 – May 2025*

- Developed firmware from scratch for a redesigned rough terrain scissor lift prototype (Mule) on a CAN-based multi-ECU off-highway vehicle architecture. Implemented platform lift, lower, and outrigger actuation using IBM Rhapsody UML statecharts and embedded C.
- Proposed and implemented a redesigned hydraulic software architecture, decomposing a monolithic handler into a dedicated statechart class with event-driven coupling, improving modularity and reducing inter-component dependencies.
- Developed the Kubota engine driver (ECON module) and built a CAN node simulator in BUSMASTER to maintain development velocity while a faulty magnetic relay was being diagnosed on the Monterrey unit prototype.
- Implemented per-wheel PID torque balancing for the hybrid drivetrain.

Diagnosed and recovered a non-functional off-highway prototype under deadline pressure: traced cascading failures (discharged battery, multiple CAN errors, erratic behavior) to a phantom load cell module incorrectly connected during factory build.
- Authored Polarion requirements for the full scissor lift family and linked them to test cases; led 30+ test case executions on physical machines, logging failures and driving fixes through full fix-verify cycles.

- Performed HIL testing using a full machine simulator, validating firmware across multiple scissor lift variants, catching integration bugs before factory deployment.
- Led cross-site knowledge transfer between Monterrey, Seattle HQ, and the Changzhou factory, collaborating with core engineering and manufacturing teams across three countries.

Embedded Software Engineer (Intern) — *John Deere* | *Monterrey, Nuevo León* | *Jul 2022*
– *Mar 2023*

- Supported firmware development for a CAN + BLE primary-secondary module pair on a forestry machine, enabling bidirectional communication between a grapple arm and the driver cabin; debugged using UART, JTAG, and direct MCU interfaces.
- Proposed and implemented a PowerShell build script improvement combining two separate firmware builds into a single step using preprocessor directives, reducing build friction for the team.

Education

B.S. in Digital Systems and Robotics Engineering | *Tecnológico de Monterrey (ITESM)* | *Nuevo León, México* | *2023*

Full Scholarship “Líderes del Mañana” – awarded for academic excellence and leadership potential.

Technical Competencies

- **Languages:** Embedded C, C++, Python, PowerShell
- **Microcontrollers:** ARM Cortex-M4, ARM Cortex-M7
- **RTOS:** embOS, MQX, freeRTOS.
- **Communication & Interfaces:** CAN, J1939, BLE, UART, RS-232
- **Tools:** J-Link, Kvaser, Peak, BUSMASTER, Git, GCC, IBM Rhapsody, Polarion, IBM Engineering Workflow Management (Jazz/RTC)
- **Practices & Methods:** V-model development, HIL testing, hardware bring-up, requirements traceability, Agile/Scrum
- **Lab Equipment:** Oscilloscope, spectrum analyzer, CAN bus sniffer, multimeter, soldering, harness building